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5CC507 Databases

Database Design I

Konstantinos Katsifis

Aims of the Session

This session aims to help you:

1. Understand the basic principles underpinning the design process of a relational database

2. Learn the basic aspects of the E/R model such as cardinality

Overview

- **Design Process**
- Example: University Database
- Entity-Relationship Model
- Cardinality

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Design Process

The design process of a database mainly involves the design of the database schema through different design phases:

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- **Conceptual-design phase**, to specify the requirements of the users, have a detailed overview of the enterprise and ensure that the requirements are met

- **Logical-design phase**, to map the high-level conceptual schema onto an implementation data model

- **Physical-design phase**, to specify the physical features of the database for the chosen DBMS

Design Process in Practice

The design process of a complex database is usually a difficult task. In practice, we split this process into different stages:

1. Capture the database requirements in the form of a *conceptual model*
2. Convert the conceptual model into a set of tables
3. Normalise the tables to eliminate *redundancy* and avoid *update anomalies*
4. Convert the tables into an implementation on a given DBMS

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E/R diagrams are the most common method for producing a conceptual model

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In the following, we will look at the first 2 stages of the design process. The other stages will be presented in a future lecture

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Example: University Database

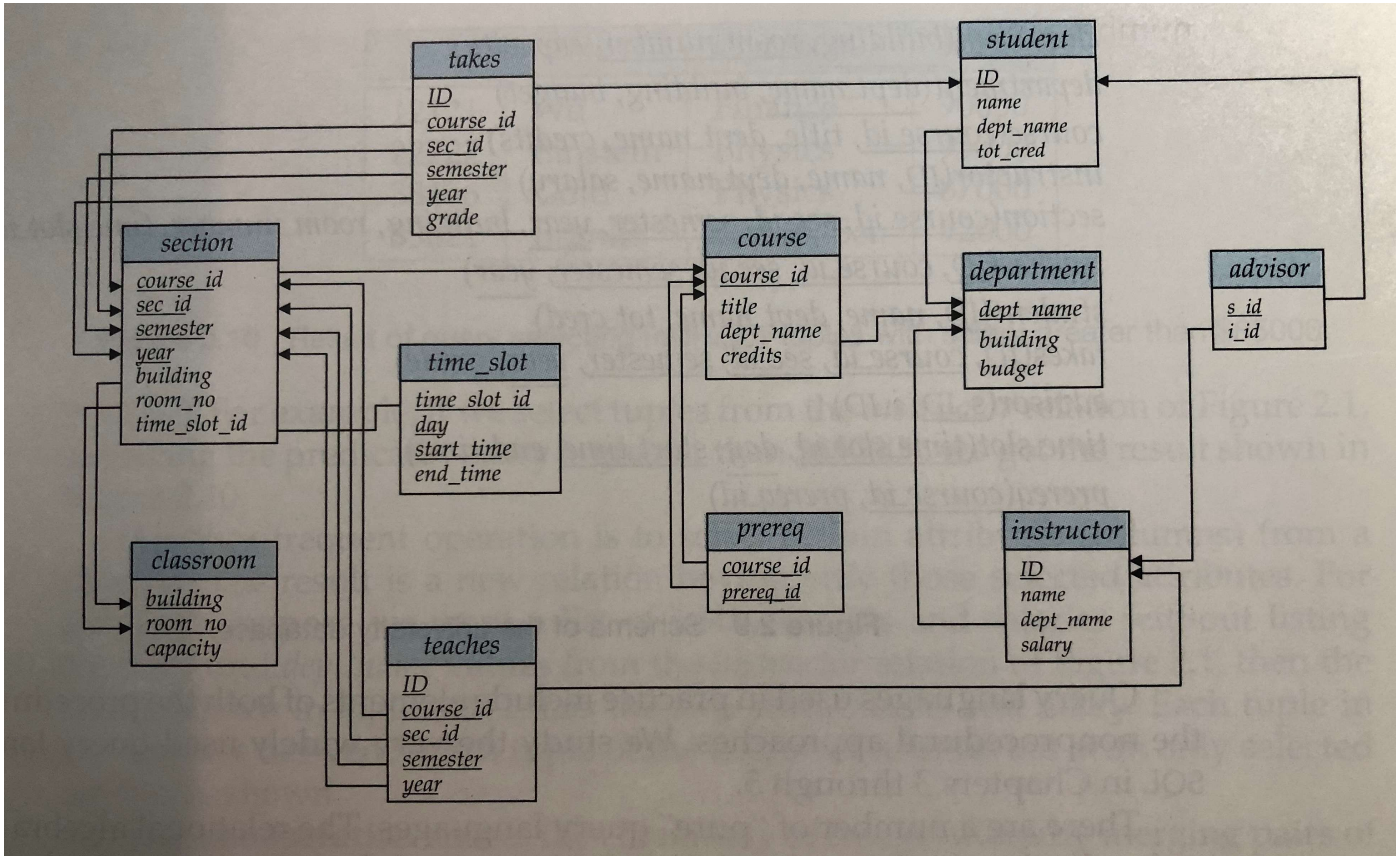

Figure 2.8 Schema diagram for the university database.

Example: University Database

Let us look at the description of the database (the tables created in Week 3):

- The university is organised into departments. Each department is identified by a unique name (*dept_name*), is located in a particular *building* and has a *budget*

department (dept_name PK, building, budget)

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Here, we identified the primary key with the acronym PK, but another way to do that is to underline the attribute

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- Students are identified by their unique *ID*. Each student has a *name*, an associated major department (*dept_name*), and the total credits the student earned so far (*tot_credit*)

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student (ID PK, name, dept_name FK, tot_credit)

Pre-Session Task

ACVITITY: In your group, take turns and share the code to create one of the tables of the database from the pre-session task with your peers. Your peers need to work out the description and the definition of this table.

Duration: 10 minutes

In the main room, you will be asked to do the same.

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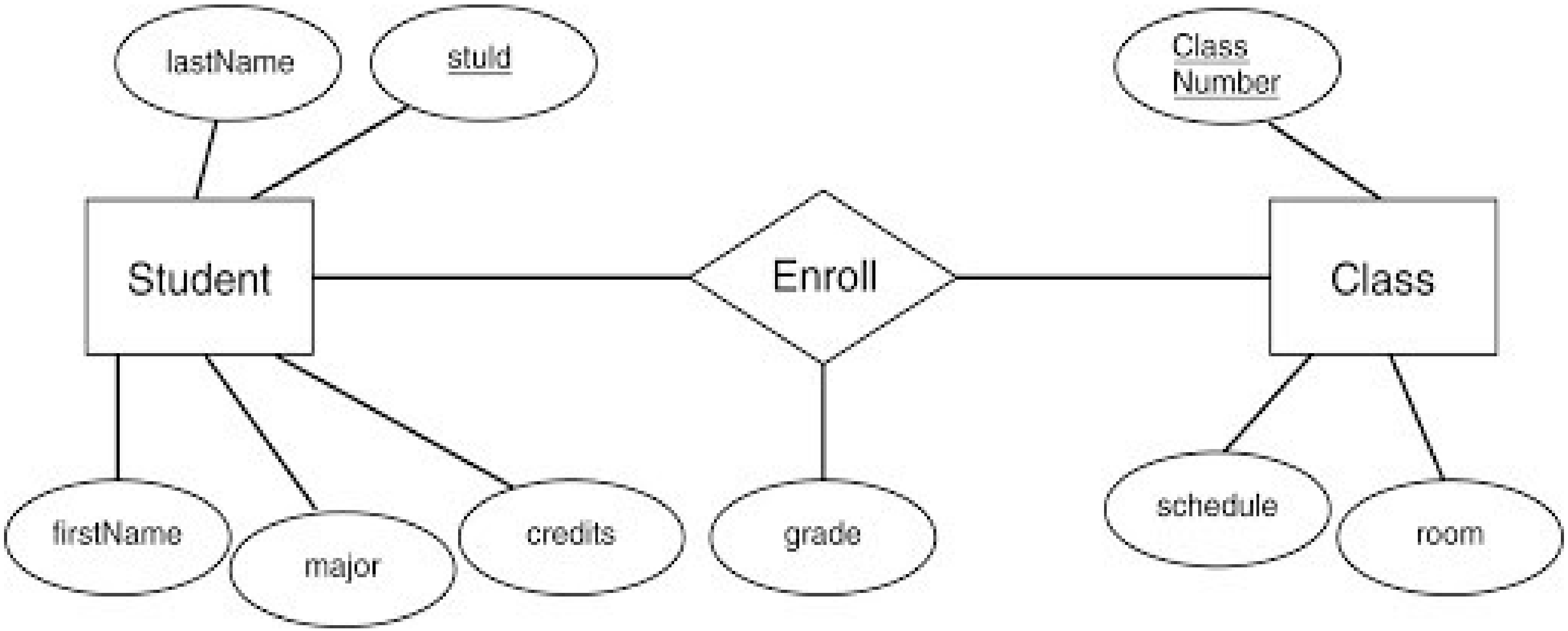
- A relationship is an association among several entities

advisor (s_id PK FK, i_id FK)

Styles of the E/R Diagram

Many different styles of diagrams, but they all show the same information



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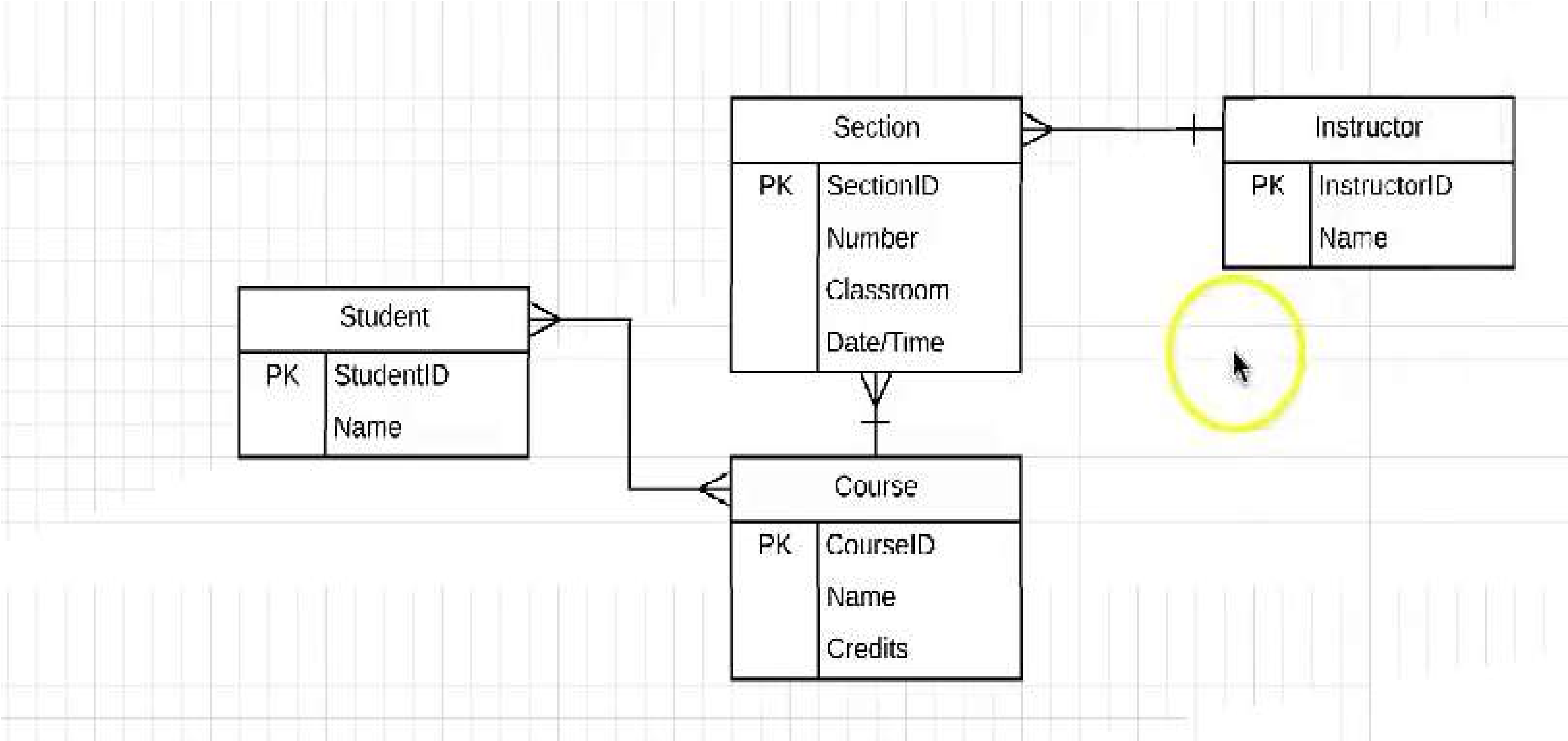

Attributes

For each attribute in an entity or relationship, there is a set of permitted values called *domain* or *value set*

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- **Derived attributes**: some attributes can be derived from other entities, e.g. *students_advised* which represents how many students each instructor advise

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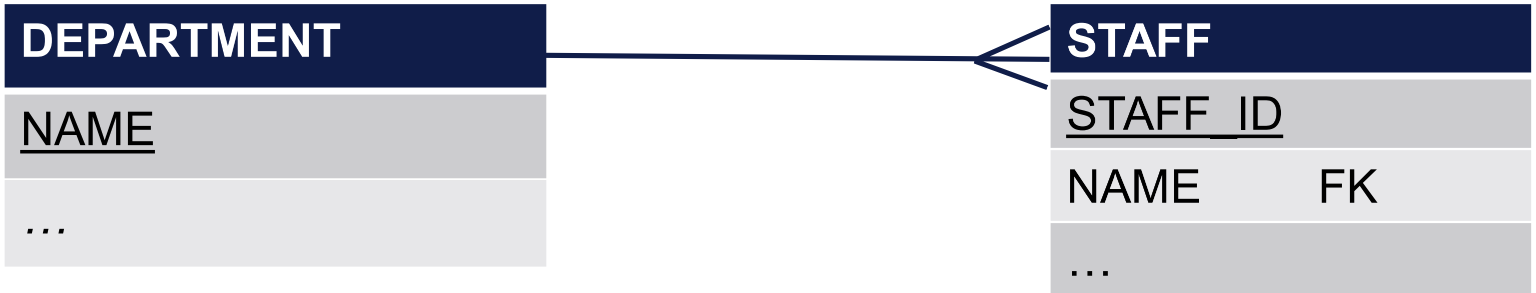
- **One to many (1:M)** – an entity in A can be associated with zero or more entities in B. An entity in B can be associated with at most one entity in A



Cardinality

Cardinality is a term used to express the number of entities to which another entity can be associated

- **Many to many (M:N)** – an entity in A can be associated with zero or more entities in B and an entity in B can be associated with zero or more entities in A

Example: University Database

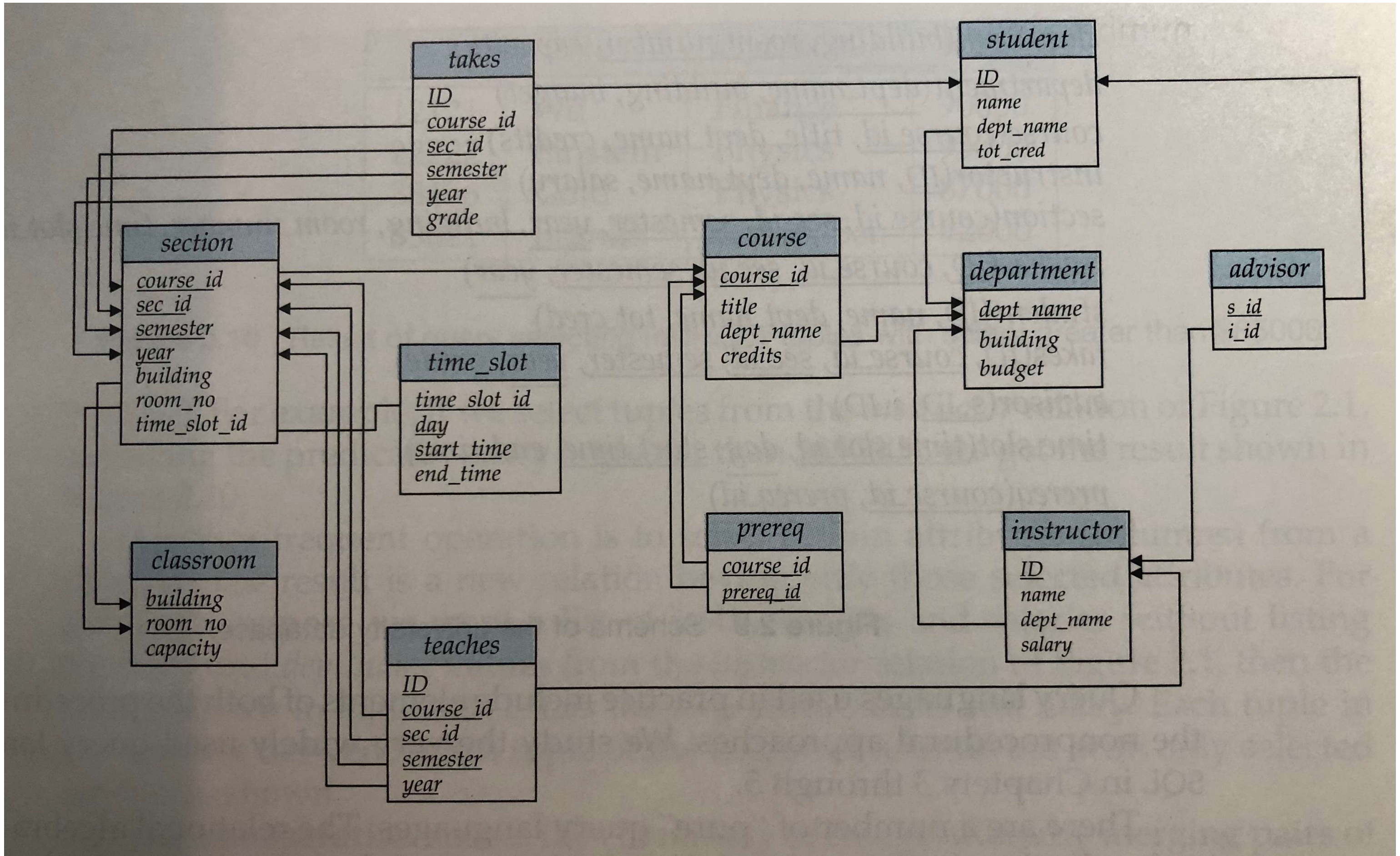


Figure 2.8 Schema diagram for the university database.

Cardinality

What is the cardinality of the following relationships?

- Instructor and Department?
- Student and Department?
- Instructor and Advisor?
- Student and Advisor?
- …

Aims of the Session

You should now be able to:

1. Understand the basic principles underpinning the design process of a relational database

2. Learn the basic aspects of the E/R model such as cardinality

References

Silberschatz, A., Korth, H. F., and Sudarshan, S. (2010), *Database System Concepts*, 6th Edition. McGraw-Hill.

- Chapter 1 – Introduction (Section 1.6 up to 1.6.4)
- Chapter 7 – Database Design and the E-R Model (up to Section 7.5.3)

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